

Long Tail Impacts of Early 2000s Subprime Loans On Today's Housing Market

M. Robson¹

¹Austin Branch
Institute of Computing in Research

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What is a Subprime Loan?

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subprime (adjective)

Subprime loans are characterized by higher interest rates, poor quality collateral, and less favorable terms in order to compensate for higher credit risk.

- Subprime loans are generally given to people with a FICO score of less than 670.
- Subprime loans usually have a rate spread of greater than 3%.

MBS and CDOs

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CDO_-_FCIC_and_IMF_Diagram.png

The 2008 Financial Crisis

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- Causes:
 - Defaults on Subprime Loans
 - Over-leveraging
 - Poor Rating Standards
- Consequences
 - Massive Financial Downturn
 - People lost their houses
 - Banks went bankrupt



Increased subprime housing expansion and credit availability.

↓ leads to

Increased production of single-family homes relative to multi-unit housing.

↓ over time leads to

Reduced long-term security due to space inefficiency and higher costs for owners and renters.

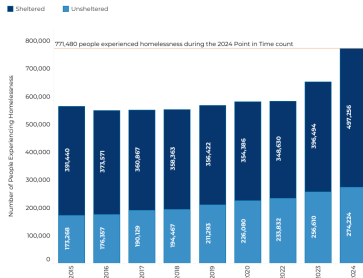
Why do we care?

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- Homelessness is increasing
- Overcrowding in homes
- Crime rate
- Poor Housing Quality
- Cost Burden is Increasing

HOMELESSNESS IS ON THE RISE



Source: U.S. Department of Housing and Urban Development, 2024 AHAR: Part 1 Point-in-Time Estimates (Accessed December 2024). <https://www.huduser.gov/portal/datasets/2024-ahar-part1-1-pt-estimates-of-homelessness-in-the-us.html>
 Note: 2020 is not included due to interruptions in data collection caused by the COVID-19 pandemic.

Data Collection

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Sources

- ACS — I used the American Community Survey (5yr) for the demographic and economic data for each tract. Additionally, this was the source for the measures of housing insecurity.
- HMDA — The Home Mortgage Disclosure Act passed allows anyone to download a list of every housing loan taken out every year, including information such as loan amount, rate spread, HOEPA status, etc.

Packages

- Polars / Pandas / GeoPandas
- Statsmodel / Bambi / Seaborn

Important Variables

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Dependent Variables

Housing Insecurity

- Cost Burden
- Overcrowding

Independent Variable

- Subprime Rate

Controlling Variables

- Demographic: Race, Undergraduate Percentage
- Economic: Income, Unemployment, Poverty Rate

Results — Overcrowding Rate

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	Baseline	(2)	(3)	(4)
Unemployment_PCT		-0.0002** (0.0001)		-0.0008*** (0.0001)
Poverty_Rate_PCT		0.0009*** (0.0000)		0.0002*** (0.0000)
Median_Household_Income		0.0001*** (0.0000)		0.0001*** (0.0000)
White_pct			-0.0013*** (0.0000)	-0.0013*** (0.0000)
Black_pct			-0.0012*** (0.0000)	-0.0011*** (0.0000)
Hispanic_pct			-0.0002*** (0.0000)	-0.0002*** (0.0000)
Undergrad_pct			-0.0004*** (0.0000)	-0.0005*** (0.0000)
Subprime_pct	0.1562*** (0.0048)	0.1029*** (0.0065)	0.0062 (0.0074)	0.0024 (0.0073)
Intercept	0.0138*** (0.0005)	0.0039*** (0.0012)	0.1438*** (0.0028)	0.1425*** (0.0030)
R-squared	0.0270	0.0581	0.3968	0.4044
Adj. R-squared	0.0270	0.0580	0.3967	0.4043
N	44,611	44,489	44,611	44,489

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

(2) Baseline + Economic, (3) Baseline + Demographic, (4) All Variables

Results — Cost Burden

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	Baseline	(2)	(3)	(4)
Unemployment_PCT		0.2010*** (0.0158)		-0.0043 (0.0156)
Poverty_Rate_PCT		-0.0059 (0.0075)		-0.2112*** (0.0082)
Median_Household_Income		0.0786*** (0.0021)		0.0266*** (0.0022)
White_pct			-0.2518*** (0.0065)	-0.2754*** (0.0056)
Black_pct			-0.1558*** (0.0071)	-0.1247*** (0.0064)
Hispanic_pct			-0.0795*** (0.0078)	-0.0678*** (0.0067)
Undergrad_pct			0.1269*** (0.0036)	0.0525*** (0.0039)
Subprime_pct	-36.8989*** (1.0786)	-12.3028*** (1.4482)	-19.8165*** (1.7683)	-25.1691*** (1.6327)
Intercept	20.2390*** (0.1329)	10.9132*** (0.3008)	32.2269*** (0.7134)	37.7072*** (0.6408)
R-squared	0.0291	0.0722	0.2389	0.2981
Adj. R-squared	0.0291	0.0721	0.2389	0.2980
N	44,716	44,489	44,716	44,489

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

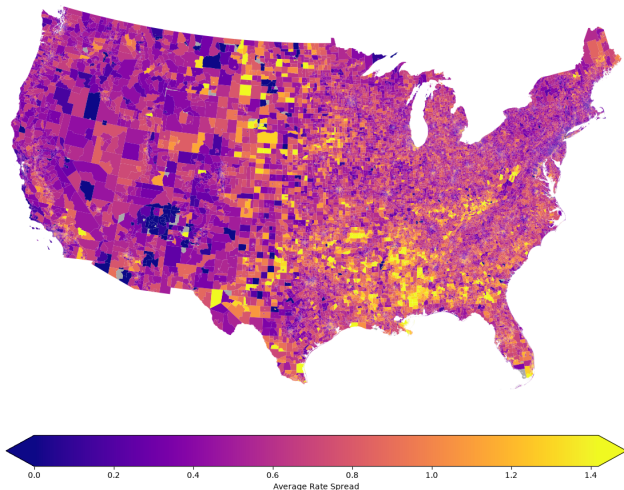
(2) Baseline + Economic, (3) Baseline + Demographic, (4) All Variables

Results — Nationwide Subprime Housing

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Average Rate Spread by US Census Tract (Continental USA)



Discussion

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Key Findings

- **Overcrowding & Demographics:** The relationship between subprime lending and overcrowding becomes statistically insignificant once demographic controls are added
- **Housing Cost Burden:** Higher historical subprime rates strongly correlate with lower current cost burdens across all models

Implications

Post-crisis price adjustments may have led to the negative association between subprime rates the cost burden. Future work could include attempt to normalize for economic status at the time of the loan creation.

Conclusion

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Other Bubbles

- AI Bubble
 - Data Center Production
 - Environmental Impacts
- Dotcom Bubble
 - Internet Infrastructure
 - Venture Capital

AI expansion is driving data center construction which could have longer term benefits to people in the future like that of the long term benefits of internet construction.